

COURSE CONTENT

UNIT I: Introduction

Importance and overview of Indian chemical industries; Unit operations and unit process concept.

A study of the following important chemical industries/ products in relation to their current status (Indian and Global), Production and consumption pattern, Manufacturing process and flow sheet, Latest technological developments, Engineering problems, viz Pollution control, Materials of construction, Corrosion, Kinetic aspects, Energy conservation and Economical status

UNIT II: Inorganic Chemical Industries

Industrial gases: Oxygen, Nitrogen, Hydrogen, Syngas,

Sulfur industry: Sulfur and sulfuric acid.

Phosphorus industry: Phosphorus, Phosphoric acid and Super phosphates.

Nitrogen industry: Ammonia, Nitric acid, Urea and other nitrogen fertilizers.

Chlor-alkali industry: Common salt, Caustic soda and Chlorine, Soda ash, Hydrochloric acid

Cement and lime

UNIT III: Organic Chemical Industries

Fermentation products: Ethanol, Acetic acid and Citric acid

Soap and detergents: Bath soap, Laundry soap, Synthetic detergent, Glycerin recovery.

Pulp and paper: Sulfate and sulfite pulps, Chemical-recovery from black liquor, Paper making, Use of recycled paper.

Paints and pigments.

Synthetic and natural fibers

TEXTBOOKS:

1. Dryden, C. E. "Outlines of Chemical Technology", 3rd Edn. (Edited and Revised by M. Gopala Rao and Sittig .M) East West Press. Pvt. Ltd, New Delhi, 3rd Edition (1997).

2. Austin G. T. "Shreve's Chemical Process Industries", 5th Edn., McGraw Hill. (1984).